HWA CHONG INSTITUTION
Preliminary Examination
Higher 2

CANDIDATE NAME	
CT GROUP	2 5 S

PHYSICS

9478 / 01

Paper 1 Multiple Choice

22 September 2026

1 hour

Additional Material: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

- Write in soft pencil.
- Do not use staples, paper clips, glue or correction fluid.
- Write your **name**, **CT group, NRIC or FIN number** on the optical mark sheet (OMS). Shade your NRIC or FIN in the spaces provided.
- There are **thirty** questions on this paper. Answer **all** questions. For each question, there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the OMS.
- Each correct answer will score one mark. A mark will **not** be deducted for a wrong answer.
- Any rough working should be done in this booklet.
- The use of an approved scientific calculated is expected, where appropriate.

Data

speed of light in free space	$c = 3.00 \times 10^{8}$ m s^{-1}
permeability of free space	$\mu_o = 4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	$\varepsilon_o = 8.85 \times 10^{-12}$ F m^{-1} $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\ \text{m F}^{-1}\right)$
elementary charge	$e = 1.60 \times 10^{-19}$ C
Planck constant	$h = 6.63 \times 10^{-34}$ J s
unified atomic mass constant	$u = 1.66 \times 10^{-27}$ kg
rest mass of electron	$m_e = 9.11 \times 10^{-31}$ kg
rest mass of proton	$m_p = 1.67 \times 10^{-27}$ kg
molar gas constant	$R = 8.31$ J K^{-1} mol^{-1}
Avogadro constant	$N_A = 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant	$k = 1.38 \times 10^{-23}$ J K^{-1}
gravitational constant	$G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}
acceleration of free fall	$g = 9.81$ m s^{-2}

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on/ by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	$T / K = T / °C + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V} < c^2 >$
mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$

displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm \omega \sqrt{\left(x_0^2 - x^2\right)}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots$
capacitors in parallel	$C = C_1 + C_2 + \ldots$
energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_0 e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_0 e^{-\lambda t}$
radioactive decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

1 The solar radiation incident on the surface of a shallow lake causes the temperature T of the lake to vary with time t. A climate model proposes that the rate of change of a lake's temperature is governed by the following equation:

$$\mu \frac{dT}{dt} = I_0 \cos(\omega t) - \kappa(T - T_G)$$

where,
I_0 is the peak power per unit area of the solar radiation on the lake,
T_G is the ground temperature, and
μ, ω, κ are constant coefficients.

What are the SI base units of the constant coefficient μ ?

A J m^{-2} K^{-1} **B** kg s^{-3} K^{-1}

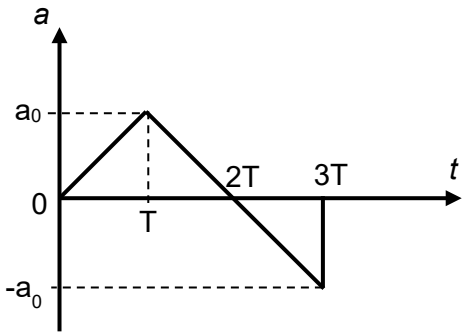
C J m^{-2} s^{-1} K^{-1} **D** kg s^{-2} K^{-1}

2 A cheetah is sprinting at 30 m s^{-1} directly towards a gazelle that is located at a direction of 50° East of North relative to the cheetah. The gazelle spots the cheetah and immediately bolts due East at 25 m s^{-1}.

What is then the magnitude of the velocity of the cheetah relative to the gazelle?

A 5.0 m s^{-1} **B** 19 m s^{-1} **C** 24 m s^{-1} **D** 52 m s^{-1}

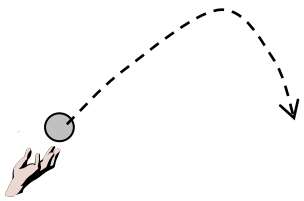
3 A car is travelling along a straight road from rest at time $t = 0$. The graph shows the variation with t of its acceleration a during part of the journey.



At which points in time does the car have its greatest velocity and greatest displacement?

	greatest velocity	greatest displacement
A	T	2T
B	2T	2T
C	2T	3T
D	3T	3T

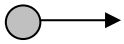
4 A ball thrown at an angle travels in a trajectory as shown.



When the ball is at the top of its flight, which of the following shows the direction of its acceleration?

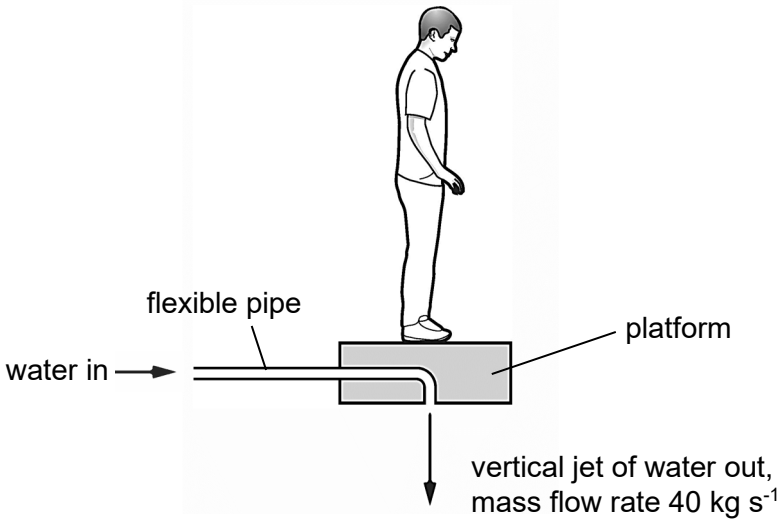
A 

B 

C 

D 

5 The diagram shows a man standing on a platform that is attached to a flexible pipe. Water is pumped through the pipe so that the man and platform remain at a constant height.



The resultant vertical force on the platform is zero. The combined mass of the man and platform is 96 kg. The mass of water that is discharged vertically downwards from the platform each second is 40 kg.

What is the speed of the water leaving the platform?

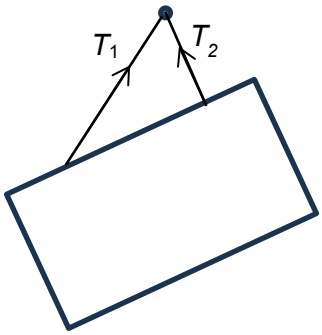
A 2.4 m s^{-1} **B** 6.9 m s^{-1} **C** 24 m s^{-1} **D** 47 m s^{-1}

6 A soccer ball of mass 0.45 kg is travelling towards a player at 3.2 m s^{-1}.

The player kicks the ball with an impulse of 6.7 N s, returning it in the direction of approach. What is the new speed of the ball?

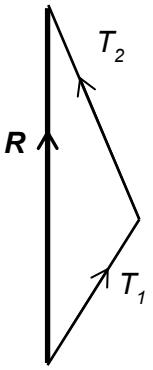
A 5.3 m s^{-1} **B** 8.1 m s^{-1} **C** 12 m s^{-1} **D** 18 m s^{-1}

7 A picture frame is hung by two cords from a nail.

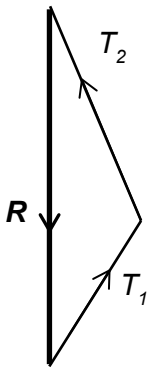


Which vector triangle represents the resultant R of the two tension forces?

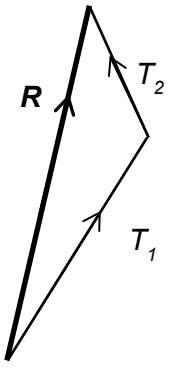
A



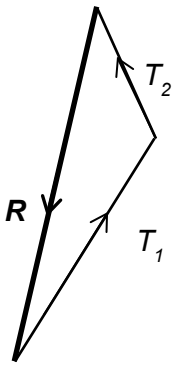
B



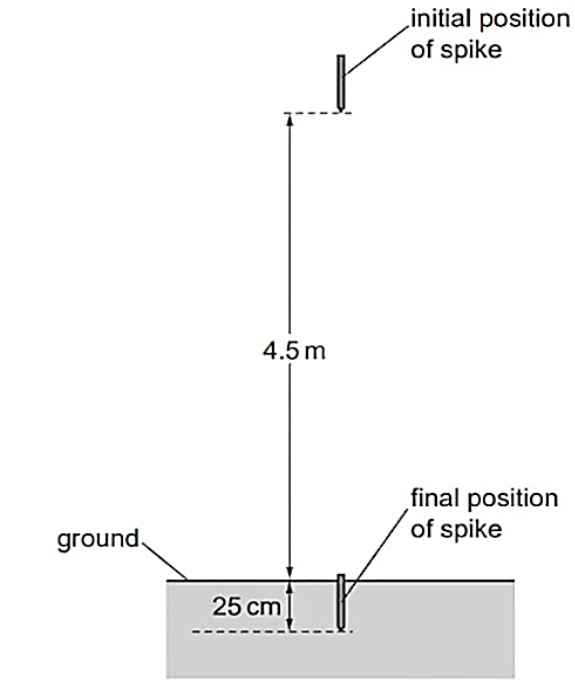
C



D



8 A metal spike of mass 3.0 kg is dropped onto level ground from a height of 4.5 m measured from the point of the spike. It penetrates into the ground a distance of 25 cm.

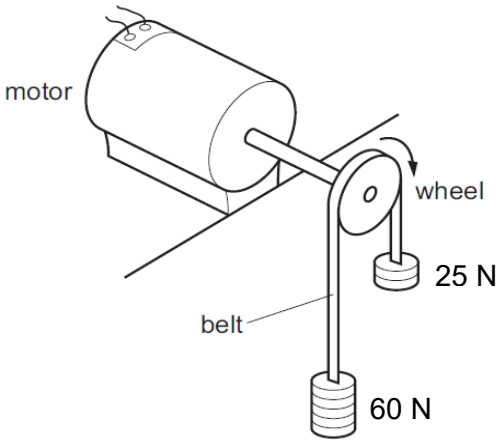


What is the mean force exerted on the spike by the ground as it penetrates?

A 5.3 N **B** 100 N **C** 530 N **D** 560 N

9 The diagram shows an arrangement used to find the output power of an electric motor.

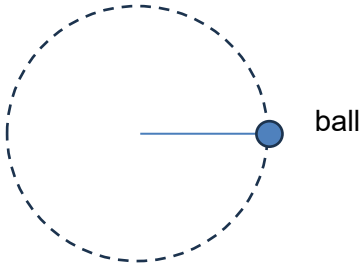
The wheel attached to the motor's axle has a diameter of 35 cm. The belt which passes over it is stationary when the wheel is making 20 revolutions per second.



What is the output power of the motor?

A 250 W **B** 770 W **C** 1300 W **D** 1900 W

10 A ball of mass 0.30 kg is attached to a light, inextensible string. It is rotated in a vertical circle. The length of the string is 0.60 m and the tangential speed of the ball when the string is horizontal is 4.0 m s^{-1}.



What is the net force acting on the ball when the string is horizontal?

A 2.9 N **B** 8.0 N **C** 8.5 N **D** 10.9 N

11 Two stars of mass *M* and 9*M* are a distance *d* apart.

Point **P** is the point where an object will experience zero resultant gravitational force due to the two stars. The distance from the centre of mass *M* to P is ***x*** and the distance from the centre of mass 9*M* to P is ***y***.

What is the ratio ***x*** **:** ***y***?

A 1 : 3 **B** 1 : 9 **C** 3 : 1 **D** 9 : 1

12 Point **Q** is a point between two planets of different masses where an object will experience zero resultant gravitational force due to the two planets.

Which statement about point **Q** is correct?

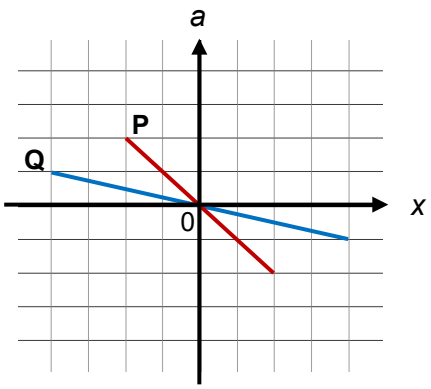
A A mass moved along the line from point **Q** to the surface of the planet of smaller mass experiences an increase in gravitational potential energy.

B The total gravitational potential at point **Q** is zero.

C The gravitational potential at point **Q** depends on the mass of any object placed there.

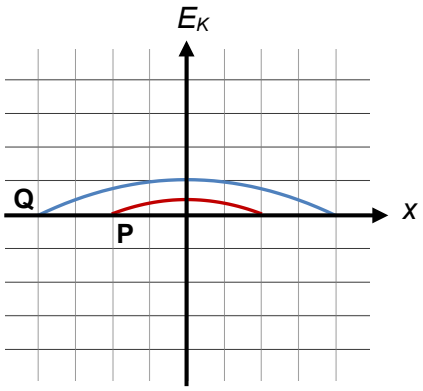
D The gravitational potential gradient at point **Q** is zero.

13 The graph shows the variation of acceleration a with displacement x for two objects, **P** and **Q** of the same mass.

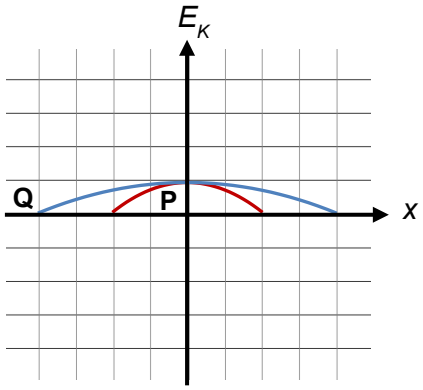


Which of the following graphs shows the corresponding variation of kinetic energy E_K with displacement x for **P** and **Q**?

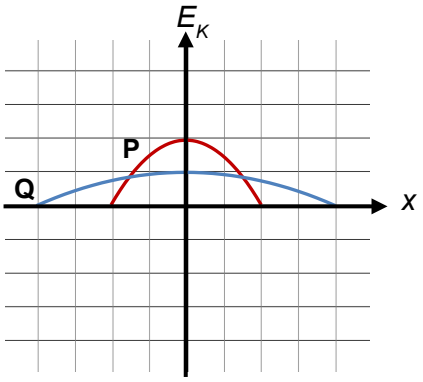
A



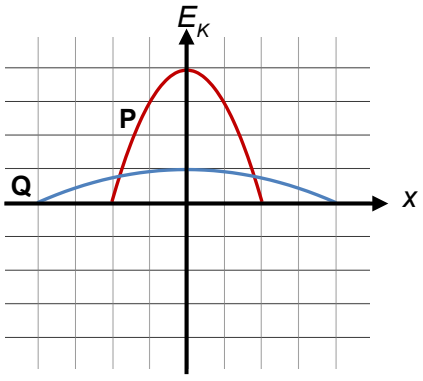
B



C



D



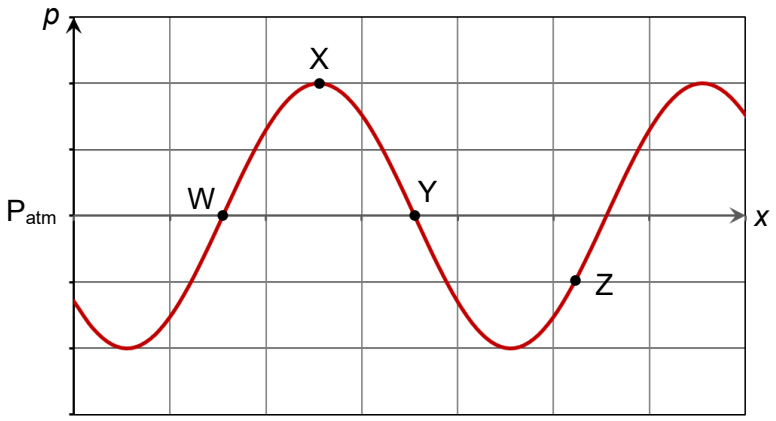
14 A mass of 0.140 kg extends a spring by 0.055 m at equilibrium. The spring has a force constant of 25.0 N m^{-1}.

The mass is pulled down a further 0.015 m from its equilibrium position and set into simple harmonic motion.

Determine the gain in elastic potential energy (EPE) of the mass when it is pulled down 0.015 m, and its energy of oscillation.

	gain in EPE / J	energy of oscillation / J
A	0.023	0.023
B	0.023	0.0028
C	0.0028	0.023
D	0.0028	0.0028

15 The graph shows the variation with position *x,* at a particular instant, of the pressure *p* in a progressive sound wave travelling through air, where P_{atm} is the atmospheric pressure.



Which of the following statements is correct?

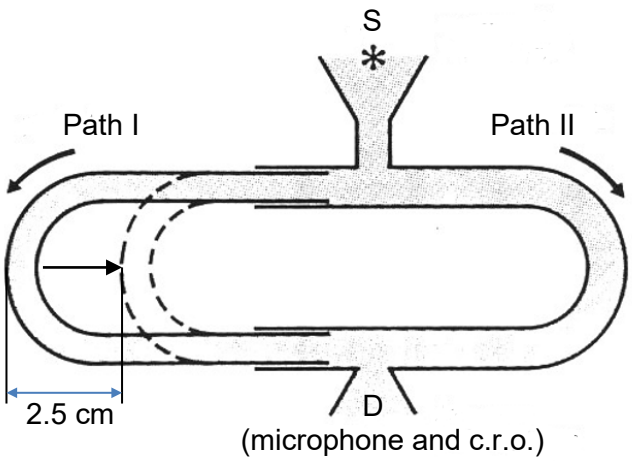
A W is a pressure node and X is a pressure antinode.

B The air particle at position X has zero displacement from its equilibrium position.

C The amplitude of oscillation of the air particle at point X is twice that of the particle at point Z.

D The phase difference between the oscillations of the air particles at point Y and point Z is $\frac{3}{2}\pi$.

16 A sound source S emits a continuous sound wave of constant frequency into a tube. The sound wave travels along two paths, I and II, before recombining at D where a microphone connected to a cathode ray oscilloscope (c.r.o) is placed.



The sliding section in Path I is adjusted until a minimum sound intensity is detected at D. The sliding section is then slowly pushed inwards. The next maximum sound intensity is detected when the sliding section has moved through a horizontal distance of 2.5 cm.

Given that the speed of sound in the tube is 340 m s^{-1}, what is the frequency of the sound wave?

A 1.7 kHz **B** 3.4 kHz **C** 6.8 kHz **D** 14 kHz

17 Two guitar strings, one 40 cm long and the other 50 cm long, are fixed at both ends under tension such that the speed of the waves is 220 m s^{-1} on both strings.

The strings are plucked simultaneously in the presence of a tuning fork of frequency 440 Hz.

Which of the two strings can cause the tuning fork to resonate?

A Neither string **B** Both strings

C The 40 cm string only **D** The 50 cm string only

18 In a heating experiment, energy is supplied at a constant rate to a liquid in a beaker of negligible heat capacity. The temperature of the liquid rises by 4.0 K per minute just before it begins to boil. After 40 minutes, all the liquid has boiled away.

For this liquid, what is the ratio of its specific heat capacity to specific latent heat of vaporisation?

A $0.10\ K^{-1}$ **B** $0.025\ K^{-1}$ **C** $0.0063\ K^{-1}$ **D** $0.0016\ K^{-1}$

19 The density of argon at a pressure of 1.00×10^5 Pa is $1.60\ kg\ m^{-3}$.

What is the root mean square speed of argon molecules at this temperature?

A $216\ m\ s^{-1}$ **B** $250\ m\ s^{-1}$ **C** $306\ m\ s^{-1}$ **D** $433\ m\ s^{-1}$

20 Two point charges of the same sign are placed 4.0 m apart. The magnitude of the charges is in the ratio 2:1. The potential at the mid-point between the two charges is 6740 V.

If the charge of smaller magnitude is changed to a negative charge of the same magnitude, calculate the electric field strength at the mid-point between the two charges.

A $6740\ N\ C^{-1}$ **B** $3370\ N\ C^{-1}$ **C** $1690\ N\ C^{-1}$ **D** $1120\ N\ C^{-1}$

21 Conducting putty is a substance that conducts electricity and can be molded into various shapes. A cylinder made from this putty has a length L and cross-sectional area A. A potential difference V is applied across its two ends. The drift velocity of the charge carriers in the putty is v_d.

This cylinder is then reshaped to one with a cross-sectional area $\frac{A}{4}$. The same potential difference V is again applied across its ends.

What is the new drift velocity of the charge carriers?

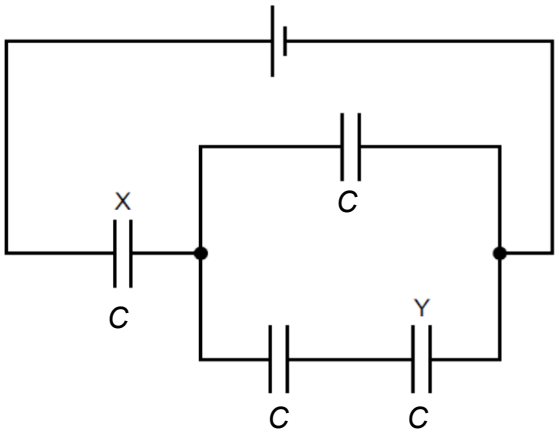
A $\frac{v_d}{4}$ **B** $\frac{v_d}{2}$ **C** v_d **D** $4v_d$

22 The capacity of an electric battery is often specified in ampere-hours.
A fully charged 12 V battery is rated 60 ampere-hours.

What is the maximum time this battery can supply power to a 36 W light bulb?

A 1.67 h **B** 3.0 h **C** 20 h **D** 180 h

23 A network of capacitors, all of the same capacitance *C*, is connected to a cell as shown.

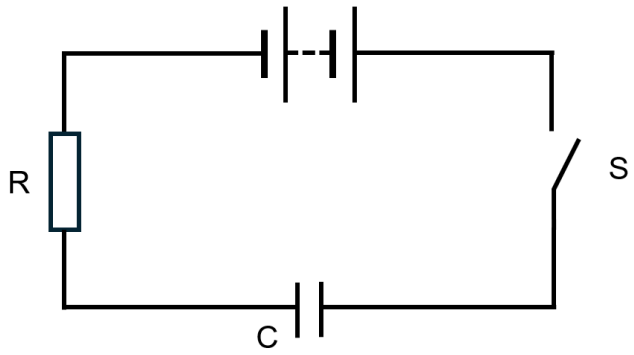


The potential difference across capacitor X is *V*.

What is the potential difference across capacitor Y?

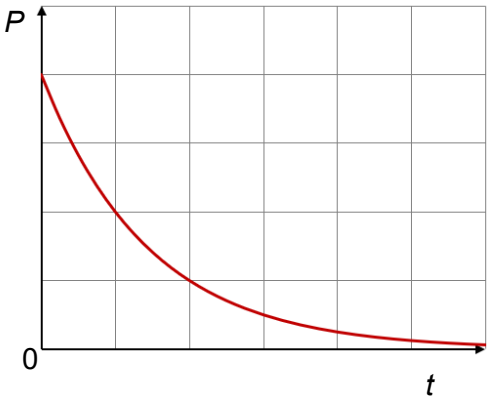
A $\frac{V}{3}$ **B** $\frac{2V}{3}$ **C** $\frac{3V}{4}$ **D** $\frac{3V}{2}$

24 An uncharged capacitor C is connected in series to a battery, resistor R and a switch S.

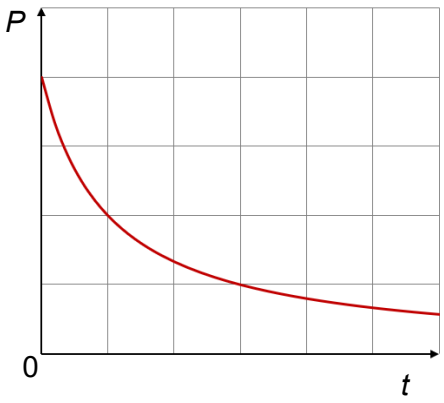


Which of the following graphs shows the variation with time *t* of the power *P* dissipated in the resistor when switch S is closed ?

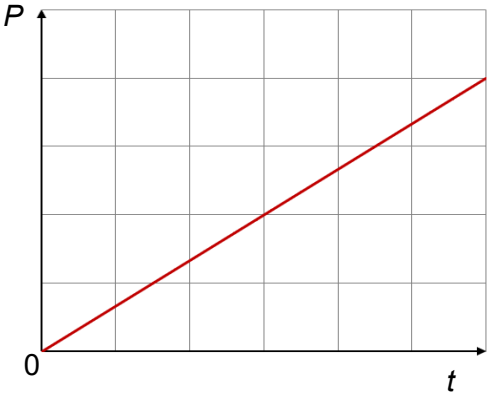
A



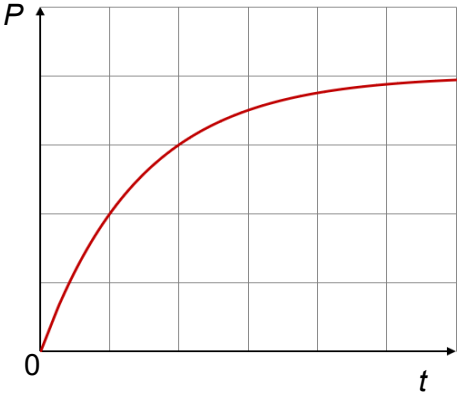
B



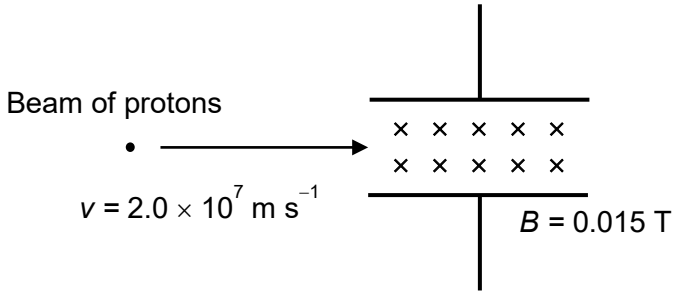
C



D



25 A beam of protons passes through a velocity selector as shown in the diagram below. The magnetic flux density, B is 0.015 T and directed into the plane of the paper



If protons travelling at 2.0×10^7 m s^{-1} pass through undeflected, what would be the direction and magnitude of the electric field?

	direction	magnitude
A	downwards	1.3×10^3 N C^{-1}
B	upwards	1.3×10^3 N C^{-1}
C	downwards	3.0×10^5 N C^{-1}
D	upwards	3.0×10^5 N C^{-1}

26 Two charged particles, X and Y, are accelerated from rest through the same potential difference V. They then enter a region of uniform magnetic field of magnetic flux density B acting perpendicular to their velocity.

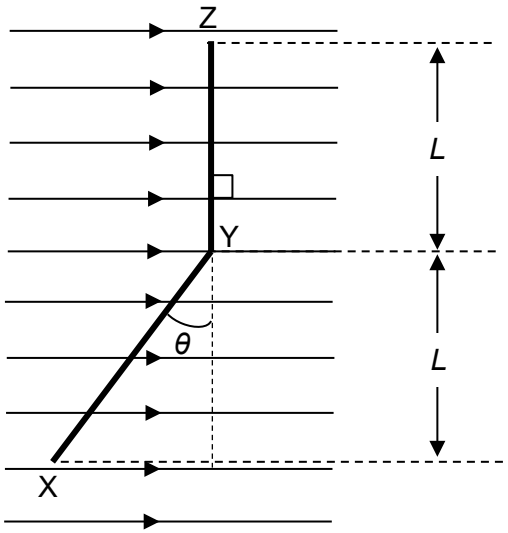
Particle X has mass m and charge q, and moves in a circular path of radius r.
Particle Y has mass $2m$ and charge $4q$.

What is the radius of the circular path of particle Y?

A $\frac{r}{2}$ **B** $\frac{r}{\sqrt{2}}$ **C** r **D** $\sqrt{2}r$

27 A rigid conductor consisting of two straight segments, XY and YZ is placed in a uniform horizontal magnetic field of flux density B as shown. A steady current I flows through the conductor from X to Y to Z.

Segment YZ has length L and is perpendicular to the magnetic field. Segment XY makes an angle θ with the vertical.

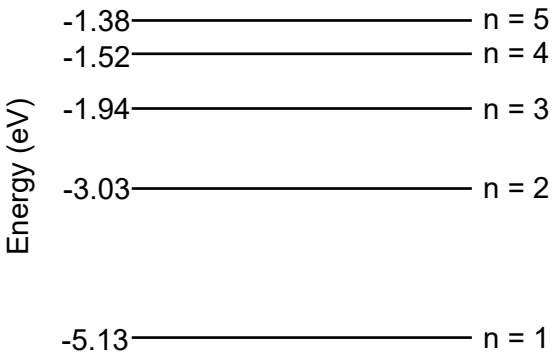


The total magnetic force experienced by the entire conductor XYZ is F.

What is the correct expression for the length of segment XY?

A $\dfrac{F}{BI\cos\theta}$ **B** $\dfrac{F}{2BI\cos\theta}$ **C** $\dfrac{F}{BI\sin\theta}$ **D** $\dfrac{F}{2BI\sin\theta}$

28 A gas has the following energy levels. It is excited by incoming electrons of kinetic energy 3.70 eV. Assume that the gas is cool and that all excitations of the gas by the incoming electrons are from the ground state.

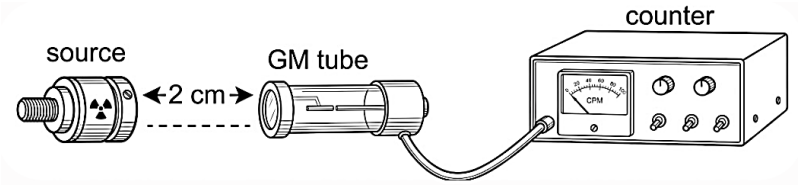


The photons emitted on subsequent de-excitation are incident on a metal with work function energy 1.50 eV.

How many emitted photons of unique wavelength are capable of causing photoelectric effect on the metal?

A 3 **B** 4 **C** 6 **D** 10

29 An experiment is set up to determine the nature of the radiation emitted by a radioactive isotope and to estimate its half-life. The source is placed 2 cm away from a Geiger-Muller (GM) tube connected to a counter. The count-rate is measured after each duration, *t*, has elapsed.



The following data is obtained:

t / hours	Count-rate with source / min^{-1}	Count-rate with source (behind a thin sheet of paper) / min^{-1}	Count-rate with source (behind a 5 mm aluminum sheet) / min^{-1}
0	3250	3247	31
24	430	431	29

Which row correctly identifies the primary type of emission and the half-life of the source?

	type of emission	half-life / hours
A	alpha	8
B	alpha	12
C	beta	8
D	beta	12

30 A sequence of radioactive decays involving three α-particles and a β-particle is shown.

$$^{237}_{93}\text{Np} \xrightarrow{\alpha} \text{Q} \xrightarrow{\beta} \text{R} \xrightarrow{\alpha} \text{S} \xrightarrow{\alpha} \text{T}$$

What is nuclide T?

A $^{225}_{88}\text{Ra}$ **B** $^{231}_{88}\text{Ra}$ **C** $^{225}_{90}\text{Th}$ **D** $^{229}_{90}\text{Th}$

~ End of Paper ~

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